Roll No. | | | | | | | |

Paper Code: TCE 499

B.Tech (Civil Engineering)
End Semester Examination 2025
IV *Semester*
Basic Geographical Information Systems

Time: Three Hours MM: 100

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions in each main question.
(iii) Total marks for each main question is **Twenty**

Q1 (10X 2 = 20 Marks)

a) Describe the comparison of GIS and Hardcopy Maps?
b) Explain the History and development of GIS world wide? CO1
c) Define the term GIS as per Good child, Burrough and ESRI?

Q2 (10X 2 = 20 Marks)

a) Explain database? "There are three ways of storing the data in a database" explain them all
b) Describe Data representation the variables of the Data representation? CO2
c) "The data for GIS collected from different sources" explain atleast three of them?

Q3 (10X 2 = 20 Marks)

a) Explain the Raster Data Models with its advantages?
b) Describe Digital Data and Cell by Cell Entry?
c) Explain the term Delaunay Triangulation with its key concepts? CO3

Q4 (10X 2 = 20 Marks)

a) Describe Vector data Sructure and Point Dictionary of Vector Data Structure?
b) "All data is analogue form need to be converted to digital form before they can be input into GIS. For methods widely used for the purpose" describe the methods od data input? CO4
c) Explain Topological Consistency and Euler Equation?

Q5 (10X 2 = 20 Marks)

a) Describe the Common uses of DEMs?
b) Explain DEM with its types, TIN and DTM? CO5
c) Describe the Methods for obtaining elevation data used to create DEMs?